

Khidmat Ghar (خدمت گار)

Software Engineering Project Documentation — Circular Reuse & Need-Based Sharing Platform

Project Parameter	Project Details
System Name	Khidmat Ghar (خدمت گار)
Platform Tagline	"Aap ke liye extra. Kisi aur ke liye zaroori." (Extra for you. Essential for someone else.)
Course & Domain	Software Engineering (BS Computer Science / Software Engineering)
Target Architecture	Web Application (HTML5, CSS3, Vanilla JavaScript, Local Storage)
Core Mission	Need-based academic donation & zero-cash peer-to-peer barter exchange

1. Project Title & System Overview

Khidmat Ghar (خدمت گار) is a web-based circular economy platform designed for academic and urban communities in Pakistan. The platform enables citizens and students to share surplus usable items through two structured modalities: (1) Need-Based Donations for genuine students, and (2) Zero-Cash Barter Exchanges between peers. The system operates on the foundational rule that 'Donation is not a free-for-all marketplace', enforcing an active claim quota (maximum 2 requests) and two-way verification to eliminate commercial hoarding and ghosting.

2. Project Background

In urban Pakistani centers such as Islamabad and Rawalpindi, households and senior students accumulate significant quantities of usable academic resources—including engineering textbooks, scientific calculators, computer peripherals, and study furniture. Simultaneously, steep inflation and textbook price increases (often Rs. 6,000 to Rs. 10,000 per curriculum book) place acute financial pressure on undergraduate students and low-income families. Due to a lack of dedicated platforms, usable items either sit idle in storage or end up as landfill waste.

3. Problem Statement

Current avenues for donating and trading used goods suffer from three core software and process failures:

- **1. Commercial Resale Hoarding (The Classifieds Problem):** On general marketplaces (OLX, Facebook), items listed for free are immediately claimed by commercial scrap dealers and flippers for resale. Genuinely needy students rarely receive these resources.
- **2. High Friction in Traditional Charities:** Traditional NGOs focus on cash donations and bulk collection, making direct peer-to-peer item donation slow, bureaucratic, and socially stigmatizing for students.
- **3. Barter Inefficiency & Trust Deficits:** Students attempting direct barter face the 'Double Coincidence of Wants'—finding a partner who both has what they need and wants what they offer—leading to endless negotiations, no-shows, and safety concerns.

4. Project Objectives

- **• Anti-Hoarding Donation Gatekeeper:** Limit each applicant to a maximum of 2 active requests with mandatory reason and impact statements.
- **• Smart Barter Matching:** Provide compatibility matching based on category, subject syllabus, distance, and user reputation without monetary exchange.
- **• 2-Way Handover OTP Verification:** Require on-site 6-digit code verification between donor and recipient to ensure physical inspection and prevent ghosting.
- **• Safe Public Meetup Integration:** Protect privacy by keeping home addresses confidential and routing transactions through 14 designated Twin Cities safe public zones.
- **• Measurable Community Impact:** Track academic financial savings (PKR) and environmental waste diversion to encourage community participation.

5. Target Users

User Persona	Role & Description	Key System Needs
Conscientious Donor	Faculty, alumni, or citizens with surplus books/gear.	Fast listing flow; review applicant reasons; verify handoff with 6-digit PIN.
Deserving Student	Undergraduates or school pupils needing course materials.	Dignified search; submit need statement; receive items safely within quota.
Smart Barter Exchanger	Students seeking to swap items without spending money.	Automated barter match score; propose 1:1 or 2-for-1 trades.
Community Moderator	Admin ensuring trust and compliance.	Audit reported listings; monitor safe zones; verify student IDs for tech aid.

6. Stakeholders and Their Roles

Stakeholder	Relationship	Primary Interest
University Students	Primary Users	Affordable access to expensive academic textbooks, tools, and calculators.
Donors & Alumni	Resource Providers	Assurance that donated goods reach sincere students rather than commercial flippers.
Universities (NUST, FAST, COMSATS, QAU)	Institutional Partners	Student financial relief, campus sustainability, and student welfare support.
Civic & Helpline Services	Safety Partners	Safe meetup routing; emergency helpline visibility (1121 Punjab, 1099 National).
Community & Municipal Bodies	Environmental Stakeholders	Reduction in municipal paper waste, e-waste, and promotion of circular reuse.

7. Functional Requirements

Req ID	Module	Functional Requirement Description
FR-01	User Profile & Quota	The system shall maintain user profiles and track an active claim quota (maximum 2 pending donation requests per user).
FR-02	Reliability Rating	The system shall record user reliability ratings (1.0 to 5.0) and award +0.1 reputation upon each verified transaction.
FR-03	Listing Creation	The system shall allow users to create listings, choosing between 'DONATE' and 'EXCHANGE' modes with category, condition, and photos.
FR-04	Faceted Catalog Search	The system shall provide search and filtering by mode (All/Donate/Exchange), category, and radius (< 5 km, < 15 km, Twin Cities).
FR-05	Privacy Fuzzing	The system shall display approximate sector locations (e.g., F-7/2 Islamabad) and never reveal exact private home addresses.
FR-06	Donation Request Form	The system shall require applicants to provide a Reason for Need and Expected Impact statement before submitting a claim.
FR-07	Donor Review & Lock	The donor shall review applicant statements and select a recipient, automatically locking the item to 'RESERVED' status.
FR-08	Smart Barter Matcher	The system shall calculate a compatibility score (up to 100%) between offered and requested items based on category, syllabus, and location.
FR-09	2-Way Handover OTP	The system shall generate a 6-digit OTP and QR code for the recipient, which the donor must enter to mark the transaction 'COMPLETED'.
FR-10	Safe Meetup Directory	The system shall list vetted public safe hubs (university gates, parks, metro stations) with safety tags and daylight operating hours.

Req ID	Module	Functional Requirement Description
FR-11	Impact Analytics	The system shall display aggregate savings (PKR), items reused, landfill waste diverted, and an interactive impact calculator.

8. Non-Functional Requirements

Quality Attribute	Requirement Specification
Usability	Clean, minimalist interface adhering to modern design standards with clear visual hierarchy, large focal numbers, and minimal text clutter.
Responsiveness	Full mobile, tablet, and desktop adaptability. Mobile layouts feature touch-friendly controls and sticky navigation.
Privacy by Design	No personal phone numbers or exact residential street addresses are rendered publicly in the catalog.
Performance	Fast client-side filtering and instantaneous score calculations with page load times under 2 seconds.
Reliability	Client-side state persistence using browser localStorage so active claims and listings survive page reloads.
Accessibility	High text-to-background contrast ratios (WCAG 2.1 AA compliant) and bilingual Urdu/English slogan representation.

9. Main System Features

- **1. Explore Catalog & Faceted Filtering:** A clean grid of circulating items with mode badges (Donate, Exchange, L2 Verified Aid), distance pills, and quick request actions.
- **2. Smart Barter Matcher (ذہین تبادلہ):** A 3-column trade arena displaying the user's item, an animated 94% compatibility ring with 4-factor progress bars, and the matched peer's textbook.
- **3. Two-Way Handover Handshake (دو طرفہ تصدیق):** A dual verification terminal showing the recipient's 6-digit code (784-219) & QR code alongside the donor's 6-box numeric PIN input.
- **4. Verified Safe Meetup Hubs (محفوظ ملاقاتی مراکز):** A transit map and directory of 14 vetted Twin Cities public meeting points with security tags (guards, CCTV) and the 4 Golden Safety Rules.
- **5. Community Impact Dashboard (کمیونٹی اثرات):** Real-time metrics (Rs. 4.8M+ saved, 1,482 items reused, 14.2 tons diverted), campus leaderboard (NUST, FAST, COMSATS), and an interactive Rupee/CO2 savings calculator.
- **6. Active Quota & Donation Gatekeeper Modal:** An interactive application popup with a visual quota meter (1/2 used) preventing hoarding and enforcing verified need statements.

10. Basic User/System Workflows

10.1 Need-Based Donation Workflow

Step	Actor	Action Taken	System Response
1	Donor	Submits item details and selects 'DONATE'.	Item is listed as AVAILABLE in the catalog with fuzzed sector location.
2	Student	Discovers item and clicks 'Request Item'.	System checks quota (< 2 active). Opens Donation Application Modal.
3	Student	Submits Reason of Need and Expected Impact.	System records pending claim and increments user active quota.
4	Donor	Reviews student reasons and clicks 'Accept'.	Item state changes to RESERVED; system generates 6-digit OTP for student.
5	Both	Meet at verified Safe Hub; student inspects item.	Student reveals 6-digit OTP (784-219) only after physical inspection.

Step	Actor	Action Taken	System Response
6	Donor	Enters 6-digit PIN in Validator Terminal.	System marks transaction COMPLETED, delists item, and awards +0.1 reputation.

10.2 Peer-to-Peer Barter Exchange Workflow

Step	Actor	Action Taken	System Response
1	User A	Lists Casio Calculator and specifies wanted CS books.	System registers barter preferences.
2	User B	Lists Discrete Math Book and seeks a calculator.	System identifies bilateral swap correlation.
3	System	Evaluates category, syllabus, location, and trust.	Calculates a 94% Symmetric Fit score on the Smart Matcher.
4	User A	Clicks 'Propose Direct 1:1 Swap'.	Dispatches trade proposal to User B.
5	User B	Accepts proposal.	Both items transition to RESERVED status.
6	Both	Meet at designated Safe Hub and exchange items & OTPs.	Both listings marked COMPLETED; mutual reputation updated.

11. Prototype Screenshots & Screen Explanations

The high-fidelity prototype was designed following modern, clean UI standards using HTML, CSS, and Vanilla JavaScript. Below are the six primary prototype screens with functional explanations:

Screen 1: Main Explore Catalog & Feed

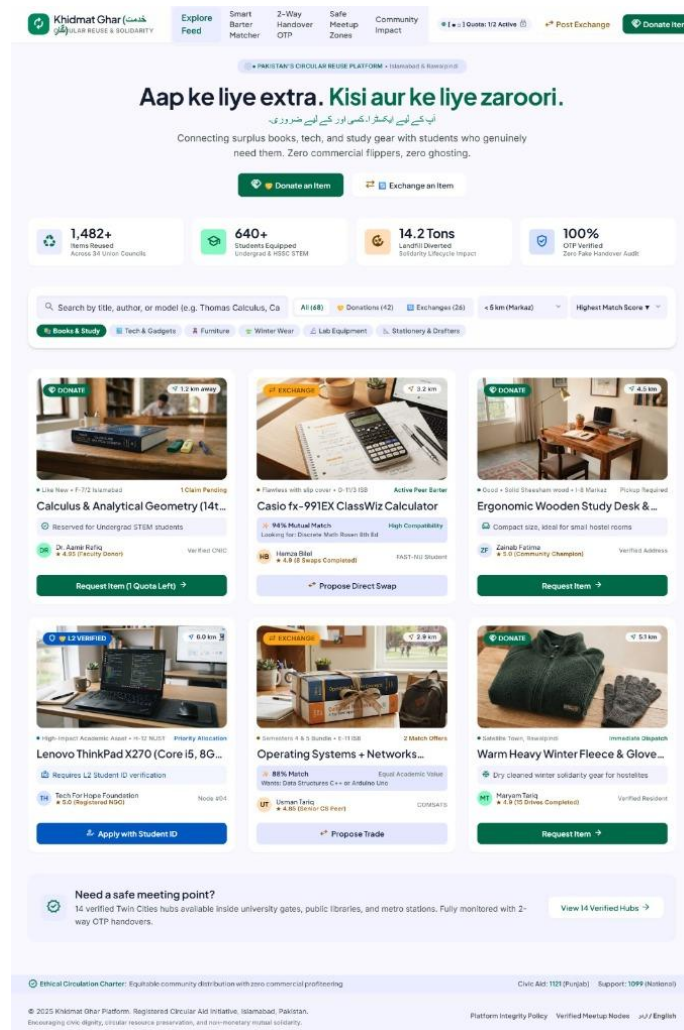


Figure 1: Explore Feed with Hero banner, impact pills, search bar, and item cards.

Explanation: This screen serves as the primary discovery hub. Users can browse items filtered by mode (Donations vs Exchanges), search by course title, and adjust radius proximity (< 5 km). Each card clearly indicates item condition, location, owner rating, and primary action buttons.

Screen 2: Smart Barter Matcher (ذہین تبادلہ)

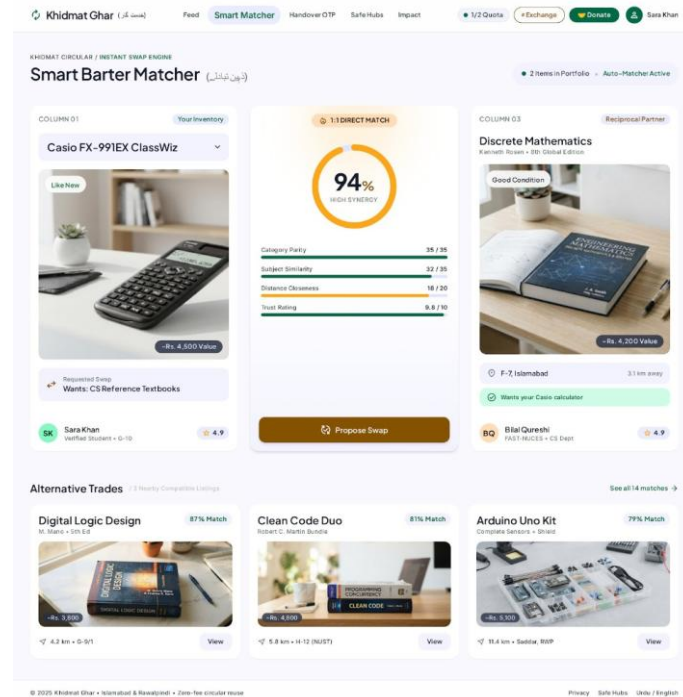


Figure 2: 3-column Smart Barter Matcher with 94% compatibility ring.

Explanation: This screen algorithmically solves the barter deadlock. It compares the user's listed item against available partner listings, displaying a 94% compatibility gauge and 4-factor breakdown bars. Users can propose direct 1:1 swaps or inspect alternative trade recommendations below.

Screen 3: Two-Way Handover OTP Protocol

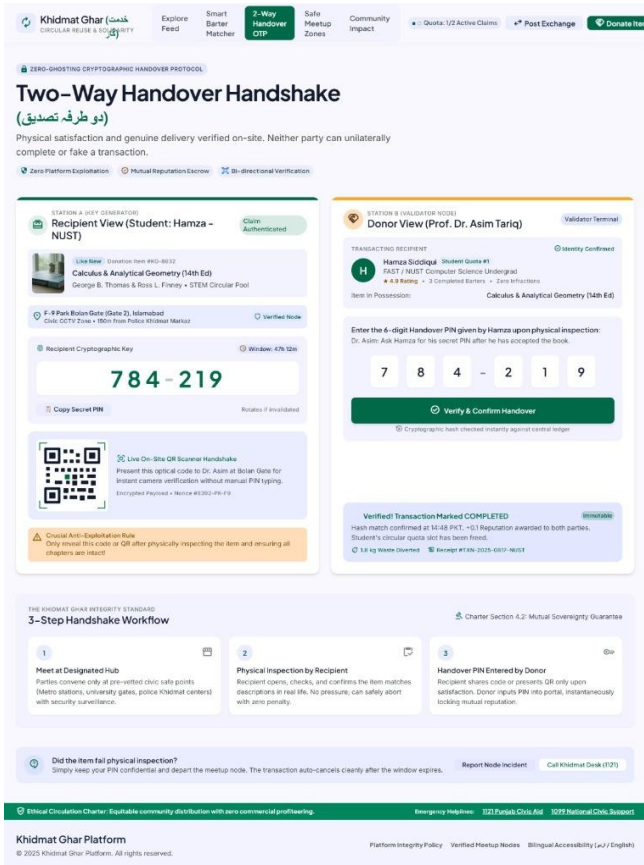


Figure 3: Recipient 6-digit PIN display and Donor Validator Terminal.

Explanation: This screen facilitates safe physical handoffs. The student recipient views a large 6-digit OTP ('784-219') and QR code to share only after physically inspecting the item. The donor inputs the code into their validator terminal to atomically finalize the transaction.

Screen 4: Verified Safe Meetup Zones

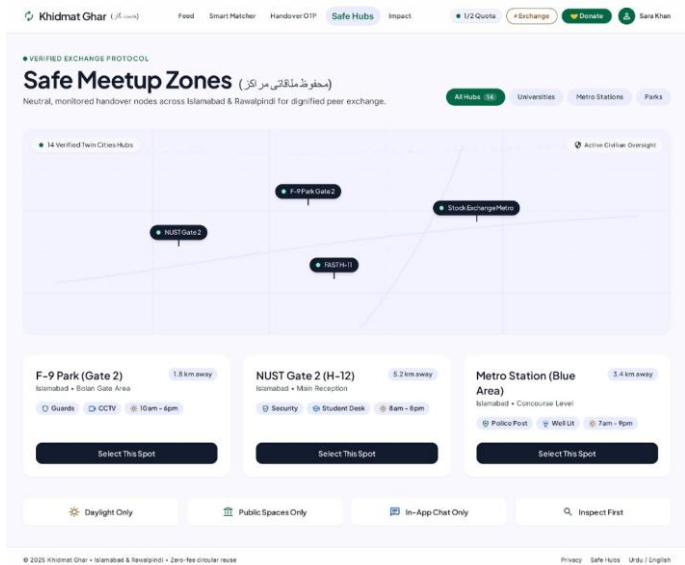


Figure 4: Public transit corridor map and verified Twin Cities safe hub directory.

Explanation: This screen provides a directory of 14 vetted public meeting spots (such as F-9 Park Gate 2, NUST Gate 2, and Metro Stations) to prevent home visits. Each location features security amenities (CCTV, guards, daylight timing) and reinforces the 4 Golden Rules of peer safety.

Screen 5: Community Impact Dashboard

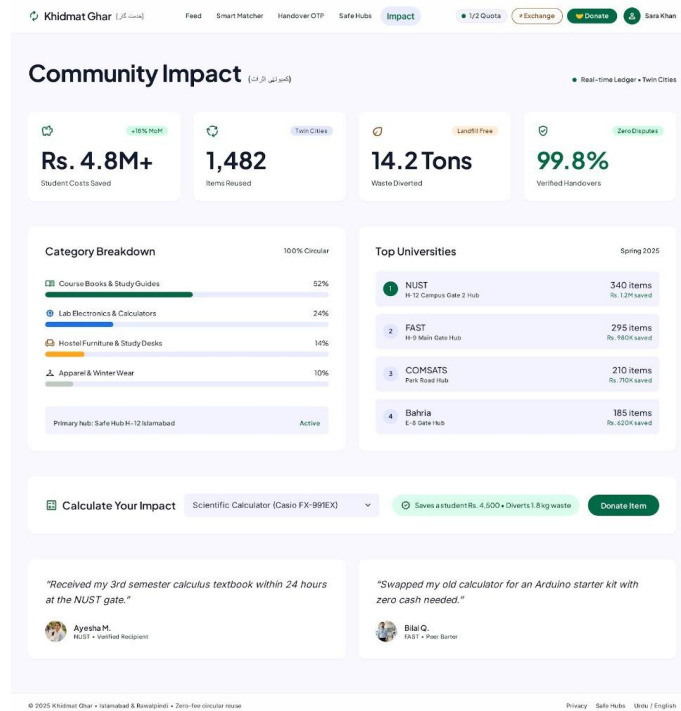


Figure 5: Community Impact metrics, campus rankings, and circular calculator.

Explanation: This dashboard showcases transparent social and ecological metrics: Rs. 4.8M+ in student costs saved, 1,482 items reused, and 14.2 tons of waste diverted. It includes an interactive calculator allowing citizens to estimate the direct monetary and carbon impact of donating their unused items.

Screen 6: Need-Based Donation Claim Modal

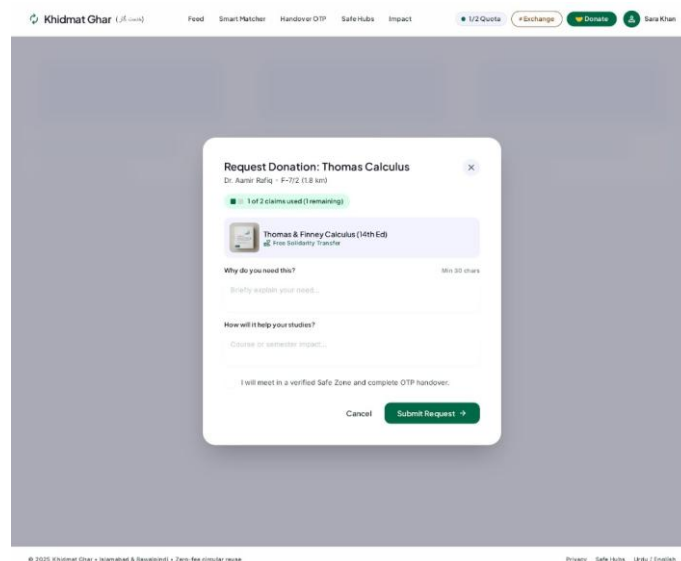


Figure 6: Donation request popup enforcing active quota and need statements.

Explanation: This modal enforces the core anti-hoarding rule when claiming an item. It displays the user's active quota status ('1 of 2 claims used') and requires concise Reason of Need and Expected Impact statements before the donor can review the request.

12. Conclusion

Khidmat Ghar demonstrates a focused Software Engineering solution to resource scarcity and waste in academic communities. By replacing the unmanaged 'free marketplace' model with need-based quota governance, multi-factor barter matching, and mutual 2-way OTP verification, the platform establishes an ethical, secure, and circular sharing ecosystem. The web prototype proves that a high-empathy, functional system can be effectively built using clean, accessible web standards (HTML, CSS, and Vanilla JavaScript).